

COMPLETE PEPTIDE REFERENCE

2025–2026 Edition | 24 Compounds

A comprehensive reference covering healing peptides, GH secretagogues, weight loss compounds, sexual health peptides, cognitive enhancers, mitochondrial agents, and next-generation metabolic research compounds. Dosing, protocols, mechanisms, side effects, and stacking strategies.

BPC-157	Healing Peptide	Accelerates healing of tendons, ligaments, and muscle tears
TB-500	Healing / Recovery Peptide	Systemic healing — works on muscles, tendons, ligaments, skin, eyes
CJC-1295 + Ipamorelin	GH Secretagogue Stack	Significantly increased GH and IGF-1 levels
Sermorelin	GH Secretagogue	Stimulates pituitary GH release without suppressing natural axis
GHRP-6	GH Secretagogue (GHRP)	Potent GH pulse stimulation — one of the strongest GHRPs
Hexarelin	GH Secretagogue (GHRP)	Strongest GH pulse stimulation of all GHRPs
IGF-1 LR3	Growth Factor	Muscle hyperplasia — stimulates satellite cell proliferation and new muscle fiber
PT-141	Sexual Function Peptide	Dramatically increases sexual desire, motivation, and libido in men and women
Melanotan II	Melanocortin Peptide	Accelerated skin tanning — deeper tan with less UV exposure required
Retatrutide	GLP-1 Triple Agonist	24.2% body weight reduction in Phase 3 (highest dose, 48 weeks) — unprecedented
Semaglutide	GLP-1 Receptor Agonist	15–17% body weight reduction at 2.4 mg/week (Wegovy) in SURMOUNT trials
Tirzepatide	GLP-1 / GIP Dual Agonist	22.5% body weight reduction at 15 mg/week (SURMOUNT-1) — best of any approved drug
GHK-Cu	Anti-Aging / Skin Peptide	Stimulates collagen and elastin synthesis — improves skin firmness and elasticity
MOTS-c	MITOCHONDRIAL	Activates AMPK — the master metabolic switch governing energy utilization

MK-677	ORAL GH RELEASE	Significantly increases GH and IGF-1 (sustained, 24-hour elevation)
Tesofensine	WEIGHT LOSS	10.6% body weight reduction at 0.5 mg/day (Phase 2 TIPO-1 trial, 24 weeks)
5-Amino-1MQ	FAT LOSS	Reduces fat cell size by reactivating metabolically dormant adipocytes
GH	GROWTH HORMONE	Powerful lipolysis — promotes fat mobilization, especially visceral and abdominal
Semax	NOOTROPIC	Rapidly elevates BDNF and NGF — promotes neurogenesis and synaptic plasticity
Selank	ANXIOLYTIC	Potent anxiolytic — reduces anxiety equivalent to low-dose benzodiazepines
Oxytocin	BONDING	Intensifies orgasm — facilitates and amplifies climax in men and women
SLU-PP-332	EXERCISE MIMETIC	70% increase in treadmill endurance capacity in preclinical studies (without trauma)
BAM-15	FAT LOSS	Significant fat loss by converting stored energy to heat (thermogenesis)
Cagrilintide	WEIGHT LOSS	~25% body weight reduction in REDEFINE 1 trial (CagriSema combination at 68 weeks)

POPULAR STACKS

Injury Healing (Gold Standard)

BPC-157 500 mcg SC + TB-500 2 mg SC — 2x/week for 4–6 weeks

GH Optimization (Clean)

CJC-1295 100 mcg + Ipamorelin 200 mcg SC before bed daily

Maximum GH Pulse

CJC-1295 100 mcg + Hexarelin 100 mcg — fasted 2x daily (6-week cycle max)

Oral GH Stack

MK-677 25 mg/night + CJC-1295/Ipamorelin on injection days

Anti-Aging / Longevity

Sermorelin 300 mcg bedtime + GHK-Cu 1 mg SC daily + MOTS-c 10 mg weekly

Elite Body Recomp

CJC-1295 + Ipamorelin + IGF-1 LR3 40–60 mcg post-workout + GH 2–3 IU

Metabolic / Fat Loss

5-Amino-1MQ 150 mg + MOTS-c 10 mg/wk + Tirzepatide or Retatrutide weekly

Maximum Weight Loss

CagriSema: Cagrilintide 2.4 mg/wk + Semaglutide 2.4 mg/wk

Sexual Performance (Men)

Tadalafil 5 mg daily + PT-141 1 mg SC 90 min before activity

Sexual Performance (Women)

PT-141 1.75 mg SC 45 min before + Testosterone cream 1 mg/day + Oxytocin 20 IU SC

Cognitive / Nootropic

Semax 400 mcg SC (AM) + Selank 250 mcg SC (PM) + MK-677 25 mg bedtime

Mitochondrial / Endurance

MOTS-c 10 mg/wk + SLU-PP-332 (research dose) + 5-Amino-1MQ 150 mg/day

Tanning + Libido (MT-II)

Melanotan II 0.5 mg SC 3x/wk + PT-141 0.5 mg SC for sexual function

Anxiety + Focus Stack

Selank 250 mcg (AM) + Semax 300 mcg (midday) + Tesofensine 0.5 mg (weight loss)

BPC-157

Body Protection Compound 157

BPC-157 is a synthetic pentadecapeptide (15 amino acids) derived from a protective protein found in human gastric juice. It has profound tissue-healing properties across multiple organ systems — from tendons and ligaments to gut mucosa and brain.

KEY BENEFITS

- Accelerates healing of tendons, ligaments, and muscle tears
- Heals leaky gut, IBS, ulcers, and intestinal damage (including NSAID-induced)
- Reduces systemic inflammation — protective for joints and connective tissue
- Promotes angiogenesis (new blood vessel formation) to injured areas
- Neuroprotective — improves recovery from traumatic brain injury
- Counteracts corticosteroid-related tendon damage
- Reduces pain and swelling at injury sites within days

HOW IT WORKS

BPC-157 upregulates growth hormone receptor expression in injured tissue, promotes VEGF (vascular endothelial growth factor) synthesis, and modulates the nitric oxide system. It activates the FAK-paxillin pathway for tendon healing and influences the gut-brain axis via vagus nerve interaction.

DOSING

Injection (SC/IM)	Oral (gut focus)	Frequency	Cycle Length
-------------------	------------------	-----------	--------------

250–500 mcg

500 mcg–1 mg

1–2x/day

4–12 weeks

PROTOCOLS

- **Injury Healing:** 500 mcg SC/IM near injury site, 1–2x daily for 4–8 weeks.
- **Gut Healing:** 500 mcg–1 mg oral (capsule or dissolve in water), taken fasted, 2x daily.
- **Systemic:** 250–500 mcg SC abdomen, 1x daily maintenance.
- **Stack Option:** BPC-157 + TB-500 for systemic + local healing synergy.

SIDE EFFECTS

- Generally very well tolerated — minimal reported side effects
- Injection site irritation (mild, transient)
- Fatigue or drowsiness at high doses (rare)
- No significant endocrine disruption

■ **Not FDA approved. Research use only. Source from reputable peptide suppliers and verify purity via HPLC testing.**

TB-500

Thymosin Beta-4 Fragment

TB-500 is a synthetic fragment of Thymosin Beta-4 (Tβ4), a naturally occurring 44-amino-acid peptide found in virtually every cell in the body. It promotes systemic healing, reduces inflammation, and improves mobility — particularly in connective tissue.

KEY BENEFITS

- Systemic healing — works on muscles, tendons, ligaments, skin, eyes
- Significant reduction in inflammation and pain
- Improves flexibility and range of motion in injured tissue
- Promotes cardiomyocyte migration and cardiac repair after injury
- Accelerates wound healing and promotes skin repair
- Stimulates stem cell differentiation and migration
- Reported improvements in hair growth and re-pigmentation

HOW IT WORKS

TB-500 works primarily by binding to actin — a protein critical for cell structure and movement. This binding promotes cell migration and proliferation in injured tissue. It upregulates actin genes in healing cells and modulates

inflammatory cytokines, reducing TNF-alpha and IL-6 while promoting anti-inflammatory pathways.

DOSING

Loading (2x/wk)	Maintenance (1x/wk)	Loading Phase	Ongoing Maintenance
2–2.5 mg	1–1.5 mg	4–6 weeks	Monthly

PROTOCOLS

- **Loading Phase:** 2–2.5 mg SC/IM 2x per week for 4–6 weeks.
- **Maintenance:** 1–1.5 mg once weekly ongoing, or monthly once healed.
- **Acute Injury:** 2.5 mg 2x/week for 2–3 weeks, then reassess.
- **Stack:** TB-500 + BPC-157 = the gold standard injury healing stack.

SIDE EFFECTS

- Generally very well tolerated at therapeutic doses
- Fatigue or lethargy (common early, usually transient)
- Injection site reactions (mild)
- Theoretical concern: may promote cancer cell migration (not shown clinically at peptide doses)

■ **Not FDA approved. Banned by WADA for competitive athletes. Research use only.**

CJC-1295 + Ipamorelin

CJC-1295 / Ipamorelin Stack

CJC-1295 (a GHRH analogue) + Ipamorelin (a GHRP/ghrelin mimetic) is the most popular GH secretagogue stack in use today. Together they produce synergistic GH pulse amplification that closely mimics the body's natural pulsatile GH release — without the cortisol and prolactin spikes of older GHRPs.

KEY BENEFITS

- Significantly increased GH and IGF-1 levels
- Accelerated fat loss — especially visceral and subcutaneous
- Improved muscle recovery, body composition, and lean mass over time
- Dramatically improved deep sleep quality (GH-wave amplification during sleep)
- Anti-aging effects: improved skin elasticity, collagen production
- Increased bone density over long-term use

- Joint and connective tissue support via IGF-1 upregulation

HOW IT WORKS

CJC-1295 (DAC version) has a half-life of ~8 days, providing sustained GHRH signal. Ipamorelin acts on ghrelin receptors to create GH pulses. Together, they simultaneously amplify GH pulse amplitude (CJC) and frequency (Ipamorelin), mimicking youth-level GH secretion without suppressing the natural HPG axis.

DOSING

CJC-1295 (no DAC)	Ipamorelin	Timing (optimal)	Cycle Length
100 mcg	100–200 mcg	Bedtime	3–6 months

PROTOCOLS

- **Standard:** CJC-1295 100 mcg + Ipamorelin 100–200 mcg SC, before bed (fasted 2 hrs).
- **CJC-DAC version:** 1–2 mg CJC-DAC 1–2x/week + daily Ipamorelin 100–200 mcg.
- **2x Daily:** Add a morning dose pre-workout for accelerated body recomp.
- **Anti-Aging:** 5 days on / 2 days off cycling prevents receptor desensitization.

SIDE EFFECTS

- Water retention (mild) — especially first few weeks
- Tingling or numbness in hands (carpal tunnel-like, dose dependent)
- Temporary insulin sensitivity reduction (monitor glucose)
- Mild fatigue within 30–60 min post-injection (expected GH effect)

■ *Long-term use may cause pituitary desensitization. Cycle with breaks. Monitor IGF-1 levels.*

Sermorelin

Sermorelin Acetate (GHRH 1-29)

Sermorelin is a synthetic analogue of the first 29 amino acids of endogenous GHRH (growth hormone releasing hormone). It stimulates the pituitary to produce and release natural GH — making it the most conservative and 'natural' approach to GH optimization. Popular in TRT clinics and anti-aging medicine.

KEY BENEFITS

- Stimulates pituitary GH release without suppressing natural axis

- Improves body composition over 3–6 months (fat loss, lean mass increase)
- Significantly improves sleep quality and deep sleep architecture
- Increases energy, vitality, and sense of well-being
- Improves skin quality and collagen density
- Supports bone density long-term
- Complementary to TRT — does not affect LH/FSH like exogenous GH

HOW IT WORKS

Sermorelin binds to GHRH receptors on somatotroph cells in the anterior pituitary, stimulating natural GH synthesis and pulsatile secretion. Unlike exogenous GH, pituitary output is still regulated by somatostatin feedback — maintaining physiologic balance and reducing risks of excess GH.

DOSING

Dose per injection	Timing	Frequency	Cycle
200–500 mcg	Bedtime	Daily	3–6 months

PROTOCOLS

- **Standard:** 200–500 mcg SC before bed, fasted, daily.
- **Anti-Aging Clinic Protocol:** Often prescribed at 300 mcg/day with annual IGF-1 monitoring.
- **Stack:** Sermorelin + Ipamorelin for GHRH + GHRP dual stimulation.
- **Cycling:** 5 days on / 2 days off to maintain receptor sensitivity.

SIDE EFFECTS

- Injection site reactions (transient, mild)
- Flushing, headache (rare at therapeutic doses)
- Water retention (less than exogenous HGH)
- Tingling (carpal tunnel-like at higher doses)

■ Available via prescription through anti-aging/TRT clinics. Compounded versions vary in quality.

GHRP-6

Growth Hormone Releasing Peptide 6

GHRP-6 is a synthetic hexapeptide that stimulates GH release by acting on the ghrelin receptor. It produces large GH pulses and is known for its powerful appetite-stimulating effect — making it particularly useful in bulking or recovery phases where high caloric intake is needed. One of the original GHRPs.

KEY BENEFITS

- Potent GH pulse stimulation — one of the strongest GHRPs
- Massive appetite increase (pro for bulking, con for cutting)
- Accelerated muscle growth when combined with caloric surplus
- Improved recovery — tissue healing and reduced DOMS
- Cardioprotective effects at therapeutic doses
- Anti-inflammatory via GH-mediated IGF-1 upregulation
- Improved sleep depth at bedtime dosing

HOW IT WORKS

GHRP-6 acts as a synthetic ghrelin agonist, binding to GH secretagogue receptor (GHSR-1a) in the hypothalamus and pituitary. This triggers somatotroph cells to release stored GH in pulses. It also amplifies GHRH-mediated GH release when stacked with CJC-1295.

DOSING

Dose	Frequency	Timing	Cycle
100–300 mcg	2–3x/day	Fasted	4–12 weeks

PROTOCOLS

- **Bulking:** 200–300 mcg 3x daily (morning, pre-workout, bedtime) — embrace the hunger.
- **GH Optimization:** 100–200 mcg with CJC-1295 2x daily.
- **Bedtime Protocol:** 200 mcg before bed for sleep + GH pulse.
- **Note:** Eat 15–20 min after injection to maximize GH pulse before insulin rise.

SIDE EFFECTS

- Significant appetite increase (ghrelin effect — can be extreme)
- Cortisol and prolactin increase (unlike Ipamorelin) — monitor on long cycles
- Water retention

- Fatigue post-injection (expected GH pulse effect)

■ *Cortisol and prolactin elevation with frequent dosing — prefer Ipamorelin for cutting or long-term use.*

Hexarelin

Hexarelin (Examorelin)

Hexarelin is among the most potent synthetic GHRPs available, producing the largest GH pulses of all peptides in its class. It has unique cardioprotective properties independent of GH release, acting directly on cardiac GHS-R receptors. Requires careful cycling to prevent receptor desensitization.

KEY BENEFITS

- Strongest GH pulse stimulation of all GHRPs
- Cardioprotective — directly improves myocardial function and reduces ischemia
- Accelerated fat loss via potent GH stimulation
- Muscle growth and enhanced recovery
- Increases IGF-1 significantly
- Neuroprotective properties
- May improve cardiac output in compromised hearts

HOW IT WORKS

Hexarelin is a synthetic GHRP-6 analogue with modifications that increase receptor affinity. It stimulates GH release via GHSR-1a and also acts directly on cardiac receptors (CD36) — explaining its GH-independent cardioprotective effects. Somatostatin is the primary limiting factor on its GH pulse size.

DOSING

Dose	Frequency	Timing	Cycle
100–200 mcg	2–3x/day	Fasted	4–6 weeks max

PROTOCOLS

- **Standard:** 100–200 mcg SC 2–3x daily, fasted.
- **Cardiac Focus:** 1–2 mcg/kg 2x daily.
- **Cycle:** Max 6 weeks, then 4–6 week break to restore receptor sensitivity.
- **Stack:** Hexarelin + CJC-1295 for maximum GH amplification.

SIDE EFFECTS

- Rapid desensitization of GH release receptors (limits cycle length)
- Cortisol and prolactin elevation
- Water retention
- Increased hunger (less than GHRP-6)

■ **Most potent GHRP — shorter cycles mandatory. Cortisol and prolactin monitoring recommended on extended use.**

IGF-1 LR3

Insulin-Like Growth Factor 1 Long Arg3

IGF-1 LR3 is a modified form of Insulin-Like Growth Factor 1 with a 13-amino-acid extension that reduces binding to IGF-BPs — increasing its half-life from minutes (native IGF-1) to approximately 20–30 hours. It promotes muscle cell hyperplasia (new cell formation), not just hypertrophy — making it uniquely anabolic.

KEY BENEFITS

- Muscle hyperplasia — stimulates satellite cell proliferation and new muscle fiber formation
- Powerful nutrient partitioning — shuttles nutrients into muscle, away from fat
- Significant fat loss via IGF-1's lipolytic action
- Enhanced recovery and protein synthesis
- Synergistic with androgens and GH peptides
- Improved glucose uptake into muscle tissue
- Cognitive benefits — neuroprotective and neurogenic effects

HOW IT WORKS

IGF-1 LR3 binds IGF-1 receptors (IGF-1R) on muscle cells, activating PI3K/Akt/mTOR pathways for protein synthesis and satellite cell activation. The LR3 modification reduces IGF-BP binding by ~1000x, dramatically extending activity. It acts both systemically and locally at injection sites.

DOSING

Daily Dose	Timing	Frequency	Cycle Max
20–100 mcg	Post-workout	Daily or ED	4–6 weeks

PROTOCOLS

- **Beginner:** 20–40 mcg IM post-workout, target worked muscles.
- **Intermediate:** 60–80 mcg post-workout, rotate injection sites.
- **Advanced:** 80–100 mcg/day split into 2 injections.
- **Cycle:** 4–6 weeks max — receptor downregulation occurs with extended use.

SIDE EFFECTS

- Hypoglycemia — most common risk; eat within 15–20 min of injection
- Jaw and organ growth concerns at sustained supraphysiologic doses
- Theoretical long-term cancer promotion risk (IGF-1 is mitogenic)
- Muscle cramps (electrolyte shift)

■ **Advanced compound.** *Hypoglycemia is a real risk — always have fast carbs on hand post-injection.*

PT-141

Bremelanotide (PT-141)

PT-141 (Bremelanotide) is a synthetic melanocortin peptide that activates MC3R and MC4R receptors in the brain to drive sexual desire and arousal at the neurological level — independent of the vascular mechanism targeted by ED drugs like Viagra. FDA approved (as Vyleesi) for hypoactive sexual desire disorder (HSDD) in women. Highly effective in men and women with low libido regardless of cause.

KEY BENEFITS

- Dramatically increases sexual desire, motivation, and libido in men and women
- Promotes erections in men via central nervous system pathways
- Effective when PDE5 inhibitors fail or are insufficient alone
- Enhances arousal, sensitivity, and sexual satisfaction in women
- Works centrally — addresses psychological/neurological drivers of low libido
- Shown effective for HSDD, post-menopause low desire, hormone-related low libido
- Spontaneous — no need to time use around sexual activity (unlike PDE5i)

HOW IT WORKS

PT-141 acts on melanocortin receptors (MC3R, MC4R) in the hypothalamus, directly activating the brain's sexual arousal circuits. Unlike PDE5 inhibitors which work peripherally on blood vessels, PT-141 works at the CNS level — increasing desire and arousal rather than just facilitating erection mechanics.

DOSING

Men (SC)	Women (FDA dose)	Timing	Frequency
0.5–2 mg	1.75 mg	1–2 hrs prior	As needed

PROTOCOLS

- **Men:** Start at 0.5–1 mg SC 1–2 hrs before activity. Increase to 2 mg if needed.
- **Women:** 1.75 mg SC 45 min before activity (FDA-approved Vyleesi dose).
- **Stack (Men):** PT-141 + low-dose Tadalafil (5 mg daily) for central + peripheral coverage.
- **Max use:** No more than 8 uses per month to avoid receptor desensitization.

SIDE EFFECTS

- Nausea (most common — take with anti-nausea OTC if needed)
- Flushing and transient blood pressure changes
- Spontaneous erections in men (dose dependent)
- Headache

■ *May cause transient blood pressure elevation. Use with caution if cardiovascular disease present.*

Melanotan II

Melanotan II (MT-II)

Melanotan II is a synthetic cyclic analogue of alpha-MSH (melanocyte-stimulating hormone). It activates multiple melanocortin receptors: MC1R (skin pigmentation), MC3R/MC4R (libido, appetite, erection), and MC5R (exocrine glands). Originally developed as a sunless tanning drug, it became popular for its libido-enhancing and appetite-suppressing side effects.

KEY BENEFITS

- Accelerated skin tanning — deeper tan with less UV exposure required
- Significantly increased libido and sexual arousal in men and women
- Spontaneous erections in men (even without sexual stimulation at higher doses)
- Appetite suppression — meaningful fat loss aid
- Stimulates lipolysis via MC4R activation
- Enhanced sensitivity and arousal in women
- UV-protective pigmentation reduces sunburn risk

HOW IT WORKS

MT-II is a cyclic peptide analogue that binds MC1R, MC3R, MC4R, and MC5R with high affinity. MC1R activation on melanocytes stimulates eumelanin (dark pigment) production. MC4R activation in the hypothalamus drives sexual arousal, appetite suppression, and erection via neurological pathways.

DOSING

Starting Dose	Maintenance	Timing	Protocol
0.25–0.5 mg	0.5–1 mg	Before sun/bed	Load then taper

PROTOCOLS

- **Loading:** 0.25 mg SC daily, gradually increase by 0.25 mg every 2–3 days to 0.5–1 mg.
- **Tanning maintenance:** 0.5 mg 2–3x/week with UV exposure.
- **Libido:** 0.5–1 mg SC 1–2 hrs before sexual activity.
- **Tip:** Start low — nausea is common at higher doses and diminishes with frequency.

SIDE EFFECTS

- Nausea and facial flushing (very common, especially on loading doses)
- Spontaneous erections (unwanted in inappropriate situations)
- Darkening of existing moles — monitor carefully
- Appetite suppression (can be profound)

■ *Mole darkening and new nevus formation reported — dermatologist monitoring recommended with regular use. Not FDA approved.*

Retatrutide

Retatrutide (GLP-1/GIP/GCG Triple Agonist)

Retatrutide (LY3437943) is Eli Lilly's next-generation triple receptor agonist targeting GLP-1 (glucagon-like peptide-1), GIP (glucose-dependent insulintropic polypeptide), and GCG (glucagon) receptors simultaneously. Phase 3 trials show 24.2% body weight reduction at 48 weeks at the highest dose — surpassing all other weight loss medications ever studied. Expected FDA approval 2025–2026.

KEY BENEFITS

- 24.2% body weight reduction in Phase 3 (highest dose, 48 weeks) — unprecedented
- Dual GLP-1 + GIP action (like Tirzepatide) plus GCG receptor for metabolic boost
- Significant visceral and liver fat reduction

- Cardiovascular risk reduction — improves lipid profile, blood pressure, HbA1c
- Appetite suppression + improved satiety signaling
- Improved insulin sensitivity and glycemic control
- Liver fat reduction — potential NASH/MAFLD treatment

HOW IT WORKS

Retatrutide agonizes three distinct receptors: GLP-1R (appetite suppression, slow gastric emptying, insulin secretion), GIPR (enhanced insulin response, fat metabolism), and GCGR (increased energy expenditure, lipolysis, hepatic glucose regulation). The glucagon component drives thermogenesis and liver fat mobilization not seen with dual agonists.

DOSING

Starting Dose (wkly)	Max Studied Dose	Frequency	Protocol
2–4 mg	Up to 12 mg	Once weekly	Titrate up

PROTOCOLS

- **Starting:** 2 mg SC weekly x 4 weeks, then titrate up every 4 weeks.
- **Target dose:** 8–12 mg/week (per Phase 3 protocol) for maximum fat loss.
- **Maintenance:** Lowest effective dose that maintains weight loss.
- **Compounded:** Available from compounding pharmacies; verify purity before use.

SIDE EFFECTS

- Nausea, vomiting, diarrhea, constipation (GI class effects — typically transient)
- Hypoglycemia (especially combined with insulin secretagogues)
- Decreased appetite and caloric intake
- Muscle mass loss (mitigate with high protein + resistance training)

■ **Contraindicated in personal/family history of medullary thyroid carcinoma or MEN2. Not approved — use compounded with verified supplier.**

Semaglutide

Semaglutide (Ozempic / Wegovy)

Semaglutide is a long-acting GLP-1 receptor agonist approved by the FDA for type 2 diabetes (Ozempic) and chronic weight management (Wegovy). SELECT trial data showed a 20% reduction in major cardiovascular events in non-diabetic patients. Produces 15–17% body weight reduction at maximum doses in clinical trials.

KEY BENEFITS

- 15–17% body weight reduction at 2.4 mg/week (Wegovy) in SURMOUNT trials
- 20% reduction in cardiovascular events (SELECT trial, 2023)
- Significant HbA1c reduction in T2DM
- Reduced liver fat and improvement in NASH markers
- Blood pressure reduction (systolic -4–5 mmHg)
- Improved lipid profile (LDL, triglycerides)
- Appetite suppression and reduced food cravings

HOW IT WORKS

Semaglutide binds GLP-1 receptors in the gut (slowing gastric emptying, increasing satiety), pancreas (stimulating insulin, inhibiting glucagon), brain (appetite centers in hypothalamus and brainstem), and cardiovascular system. Its fatty acid side chain extends half-life to ~1 week via albumin binding.

DOSING

Starting Dose	Max (Wegovy)	Frequency	Protocol
0.25 mg/wk	2.4 mg/wk	Once weekly	4-week titration

PROTOCOLS

- **Ozempic (diabetes):** 0.25 mg weekly x4 → 0.5 mg x4 → 1 mg x4 → 2 mg.
- **Wegovy (obesity):** 0.25 mg x4 → 0.5 → 1.0 → 1.7 → 2.4 mg (each 4 weeks).
- **Compounded:** Slower titration to minimize GI side effects.
- **Protein:** 1g/lb body weight minimum to preserve lean mass during loss.

SIDE EFFECTS

- Nausea (most common — diminishes over weeks)
- Vomiting, diarrhea, constipation
- Gastroparesis risk with long-term use

- Muscle mass loss (offset with protein + resistance training)

■ **Contraindicated in personal/family history of MTC or MEN2. Pancreatitis — discontinue if severe abdominal pain develops.**

Tirzepatide

Tirzepatide (Mounjaro / Zepbound)

Tirzepatide is a once-weekly dual GIP/GLP-1 receptor agonist approved by the FDA for type 2 diabetes (Mounjaro) and obesity (Zepbound). SURMOUNT-1 trials demonstrated 22.5% body weight reduction at the highest dose — currently the most effective approved weight loss medication. Adding GIP agonism to GLP-1 produces synergistic metabolic effects.

KEY BENEFITS

- 22.5% body weight reduction at 15 mg/week (SURMOUNT-1) — best of any approved drug
- Superior HbA1c reduction vs. semaglutide (SURPASS trials)
- Significant visceral fat reduction
- Improved insulin sensitivity and beta-cell function
- Liver fat reduction (MAFLD/NASH improvement)
- Cardiovascular risk reduction (SURPASS-CVOT underway)
- Reduced hypertension and improved lipid profiles

HOW IT WORKS

Tirzepatide is a 'twincretin' — a single molecule that acts on both GLP-1R and GIPR with balanced potency. GIP agonism synergizes with GLP-1 to enhance insulin secretion, improve adipose tissue metabolism, and reduce glucagon. The combined receptor activation produces greater weight loss than GLP-1 alone.

DOSING

Starting Dose	Max Dose	Frequency	Titration
2.5 mg/wk	15 mg/wk	Once weekly	4-week steps

PROTOCOLS

- **Standard:** 2.5 mg x4 weeks → 5 mg x4 → 7.5 → 10 → 12.5 → 15 mg/week.
- **Aggressive recomp:** Pair with calorie deficit, 1g/lb protein, 3–4x resistance training.
- **Maintenance:** Lowest dose maintaining body composition goals.

- **Semaglutide switch:** No washout needed — titrate up from starting Tirzepatide dose.

SIDE EFFECTS

- Nausea, vomiting (less than semaglutide at comparable doses)
- Diarrhea, constipation
- Muscle mass loss (critical to offset with training + protein)
- Injection site erythema, bruising

■ **Same thyroid tumor warnings as GLP-1 class. Not for personal/family history of MTC or MEN2.**

GHK-Cu

Copper Peptide GHK-Cu

GHK-Cu (glycyl-L-histidyl-L-lysine:copper) is a naturally occurring human plasma peptide that declines significantly with age. It acts as a master regulator of tissue repair and renewal — promoting collagen synthesis, wound healing, hair follicle regeneration, and gene expression reprogramming toward a more youthful state.

KEY BENEFITS

- Stimulates collagen and elastin synthesis — improves skin firmness and elasticity
- Reduces fine lines and wrinkles (validated in multiple clinical studies)
- Promotes wound healing and scar remodeling
- Stimulates hair follicle proliferation — reduces hair loss, promotes regrowth
- Anti-inflammatory — reduces pro-inflammatory cytokines
- Antioxidant — chelates copper into protective pathways
- Reprograms aged gene expression toward younger patterns (Pickart research)

HOW IT WORKS

GHK-Cu acts as a biological 'switch' that resets aging gene expression. It upregulates over 31 genes related to tissue repair and downregulates 36 genes associated with inflammatory signaling and cancer progression. It also activates proteasomes (cellular cleanup systems) and promotes angiogenesis in damaged tissue.

DOSING

SC/IM (injectable)	Topical creams	Frequency	Duration
0.5–2 mg	Topical 1–3%	Daily	Ongoing

PROTOCOLS

- **Topical Anti-Aging:** 1–3% GHK-Cu serum applied 2x daily to face/scalp.
- **Injectable:** 0.5–2 mg SC daily or 3–5x weekly.
- **Hair Loss:** Scalp massage with GHK-Cu serum + topical minoxidil for synergy.
- **Stacks well with:** BPC-157, TB-500 for enhanced healing protocols.

SIDE EFFECTS

- Topical: mild skin redness, irritation (transient, usually at high concentrations)
- Injectable: injection site reactions (mild)
- Generally considered very safe — naturally occurring in human plasma
- Rare: skin discoloration at injection sites

■ *Use copper-peptide compatible skincare — avoid high-pH products (Vitamin C serums, retinoids) simultaneously as they degrade peptide.*

MOTS-c

Mitochondrial Open Reading Frame of the 12S rRNA-c

MOTS-c is a mitochondrial-derived peptide (MDP) encoded within the 12S ribosomal RNA gene of mitochondrial DNA. First identified in 2015 by Lee et al. at USC, it functions as a systemic metabolic regulator — improving insulin sensitivity, promoting fat utilization, enhancing mitochondrial function, and extending lifespan in animal models. It declines significantly with age, making it a compelling longevity and metabolic compound.

KEY BENEFITS

- Activates AMPK — the master metabolic switch governing energy utilization
- Improves insulin sensitivity — reduces insulin resistance in skeletal muscle
- Promotes fat oxidation and lipolysis — shifts fuel use toward stored fat
- Enhances mitochondrial biogenesis and function
- Exercise mimetic — replicates some benefits of aerobic exercise without movement
- Extends healthy lifespan in mouse models
- Anti-inflammatory — reduces oxidative stress systemically

HOW IT WORKS

MOTS-c acts in a AMPK-dependent manner to activate the folate cycle and one-carbon metabolism, leading to AICAR accumulation — a natural AMPK activator. This mirrors the metabolic effects of exercise and caloric restriction. It also translocates to the nucleus under stress, regulating nuclear gene expression and the adaptive

response to metabolic challenge.

DOSING

Weekly SC dose	Daily SC (alt. protocol)	Optimal timing	Cycle
5–15 mg	1–5 mg	Morning	8–12 weeks

PROTOCOLS

- **Weekly Protocol:** 10 mg SC once weekly, morning fasted.
- **Daily Protocol:** 2–5 mg SC daily, cycle 8 weeks on / 4 weeks off.
- **Exercise Stack:** Dose pre-training — synergistic with aerobic exercise for metabolic adaptation.
- **Longevity Stack:** MOTS-c + Semax + NMN/NR for comprehensive mitochondrial and cognitive support.

SIDE EFFECTS

- Generally well tolerated — no significant adverse effects in current human use
- Injection site reactions (mild, transient)
- Mild fatigue or lightheadedness at higher doses (AMPK activation effect)
- Hypoglycemia possible if combined with insulin or diabetes medications

■ **Research peptide — no human clinical trials completed. Source from reputable suppliers with HPLC certification.**

MK-677

Ibutamoren (MK-677) — Oral GH Secretagogue

MK-677 (Ibutamoren) is a non-peptide oral ghrelin mimetic and GH secretagogue. It stimulates GH and IGF-1 release by acting on GHSR-1a receptors — and crucially, it is orally bioavailable with a 24-hour half-life, requiring just one dose per day. The only oral compound that significantly and sustainably raises GH and IGF-1 long-term without injections. Extensively studied in clinical trials.

KEY BENEFITS

- Significantly increases GH and IGF-1 (sustained, 24-hour elevation)
- Oral — no injections required (major compliance advantage)
- Accelerates muscle growth and recovery via IGF-1 upregulation
- Dramatically improves sleep — increases REM and deep sleep stages
- Improves bone density — shown in multiple clinical trials

- Reduces body fat while preserving lean mass in caloric deficits
- Anti-catabolic — preserves muscle during periods of restriction

HOW IT WORKS

MK-677 mimics the action of ghrelin at the GHSR-1a receptor in the hypothalamus and pituitary, amplifying natural GH pulse amplitude without disrupting pulsatility. Unlike injectable GHRPs, the 24-hour half-life provides sustained IGF-1 elevation rather than discrete pulses. It does not suppress natural GH axis or affect cortisol at therapeutic doses.

DOSING

Daily oral dose	Optimal timing	Frequency	Cycle (long)
10–25 mg	Before bed	Once daily	16–24 weeks+

PROTOCOLS

- **Standard:** 25 mg orally before bed (food optional) — maximizes GH pulse during sleep.
- **Sensitive start:** Begin at 10 mg/day for 2 weeks, then increase to 25 mg.
- **Cutting:** 10–15 mg to balance IGF-1 benefits against water retention and hunger.
- **Stack:** MK-677 + CJC-1295/Ipamorelin on injection days = dual GH stimulation.

SIDE EFFECTS

- Significant appetite increase (ghrelin mimetic effect — plan calories accordingly)
- Water retention, especially first 2–4 weeks
- Temporary insulin resistance / blood glucose elevation — monitor fasting glucose
- Fatigue or lethargy (GH effect — take at bedtime to minimize)

■ **Monitor fasting glucose and HbA1c every 8–12 weeks on extended cycles. Water retention may cause temporary weight gain unrelated to fat.**

Tesofensine

Tesofensine — Triple Monoamine Reuptake Inhibitor

Tesofensine is a novel triple reuptake inhibitor of dopamine, norepinephrine, and serotonin (similar mechanism to phentermine but more selective). Originally developed for Parkinson's and Alzheimer's disease, Phase 2 weight loss trials showed 10.6% body weight reduction at 0.5 mg/day over 24 weeks — outperforming most approved anti-obesity drugs of its era. It reduces appetite through multiple central pathways simultaneously.

KEY BENEFITS

- 10.6% body weight reduction at 0.5 mg/day (Phase 2 TIPO-1 trial, 24 weeks)
- Significant reduction in appetite and food cravings
- Increases basal metabolic rate (norepinephrine effect)
- Improved energy and motivation (dopamine + norepinephrine)
- Mood enhancement (serotonin component)
- Superior to most single-mechanism appetite suppressants
- Synergistic with GLP-1 agonists for combined central + peripheral appetite control

HOW IT WORKS

Tesofensine blocks the reuptake transporters for dopamine (DAT), norepinephrine (NET), and serotonin (SERT) — increasing synaptic levels of all three monoamines. Norepinephrine increases energy expenditure and reduces appetite via beta-adrenergic receptors. Dopamine reduces reward-driven eating. Serotonin signals satiety centrally. The triple mechanism prevents receptor adaptation that limits single-mechanism drugs.

DOSING

Starting dose	Standard dose	Max dose (caution)	Timing
0.25 mg/day	0.5 mg/day	1.0 mg/day	Morning

PROTOCOLS

- **Start low:** 0.25 mg orally daily x4 weeks, then titrate to 0.5 mg.
- **Standard protocol:** 0.5 mg/day morning, assess at 8 weeks for response.
- **Cycling:** 12–16 weeks on, 4–8 weeks off to prevent monoamine depletion.
- **Stack:** Tesofensine + GLP-1 agonist (semaglutide/tirzepatide) for additive weight loss.

SIDE EFFECTS

- Dry mouth (very common — most frequently reported side effect)
- Nausea (common early, diminishes)
- Increased heart rate (norepinephrine effect — monitor cardiovascular status)
- Insomnia if taken late in day

■ **Cardiovascular risk** — *contraindicated in uncontrolled hypertension, arrhythmia, or structural heart disease. Not for use with other serotonergic drugs.*

5-Amino-1MQ

5-Amino-1-Methylquinolinium — NNMT Inhibitor

5-Amino-1MQ is a small molecule inhibitor of NNMT (nicotinamide N-methyltransferase) — an enzyme overexpressed in fat tissue that suppresses fat cell metabolism. By blocking NNMT, 5-Amino-1MQ raises cellular NAD⁺ and SAM levels, reactivates metabolically dormant fat cells, and causes adipocytes to shrink without requiring caloric restriction. Preclinical research shows significant fat loss with preservation of lean mass.

KEY BENEFITS

- Reduces fat cell size by reactivating metabolically dormant adipocytes
- Increases intracellular NAD⁺ — mimics caloric restriction at the cellular level
- Raises SAM (S-adenosylmethionine) levels — enhances methylation and epigenetic health
- Reduces adipogenesis — inhibits formation of new fat cells
- Preserves or increases lean muscle mass during fat loss
- Anti-aging properties via NAD⁺/sirtuin pathway activation
- Improved insulin sensitivity in fat tissue

HOW IT WORKS

NNMT consumes NAD⁺ precursors (specifically 1-methylnicotinamide from NAM) in fat tissue, keeping adipocytes in a metabolically suppressed state. 5-Amino-1MQ blocks this enzyme, freeing NAD⁺ precursors to restore cellular NAD⁺ levels. Higher NAD⁺ activates sirtuins and AMPK — triggering fat oxidation, mitochondrial biogenesis, and metabolic reactivation of adipose tissue.

DOSING

Daily oral dose	Timing	Frequency	Cycle
50–200 mg	Morning	Daily	8–12 weeks

PROTOCOLS

- **Starting dose:** 50–100 mg orally daily, assess response at 4 weeks.
- **Full dose:** 150–200 mg/day in split doses (AM + midday).
- **NAD⁺ Stack:** 5-Amino-1MQ + NMN 500 mg + Resveratrol 500 mg = synergistic NAD⁺ pathway.
- **Metabolic Stack:** 5-Amino-1MQ + MOTS-c + Berberine for multi-pathway fat loss.

SIDE EFFECTS

- Limited human data — primarily preclinical (rodent) research

- GI discomfort reported by some users at higher doses
- Mild headache during initial adjustment period
- Generally well tolerated in current anecdotal human use

■ **Very early-stage research compound. No completed human clinical trials. Use with appropriate caution and bloodwork monitoring.**

GH

Human Growth Hormone (Somatropin / rHGH)

Human Growth Hormone (HGH/somatropin) is a 191-amino-acid peptide hormone secreted by the anterior pituitary. Pharmaceutical GH (recombinant human growth hormone / rHGH) is identical in structure to endogenous GH. It is used clinically for GH deficiency, short stature, HIV wasting, and Turner syndrome. In performance and anti-aging medicine, it is used for body recomposition, fat loss, recovery, and longevity.

KEY BENEFITS

- Powerful lipolysis — promotes fat mobilization, especially visceral and abdominal fat
- Increases IGF-1 — drives muscle protein synthesis and recovery
- Significantly improves body composition (fat loss + lean mass over 3–6+ months)
- Enhanced recovery — reduces muscle soreness, accelerates tissue repair
- Improved sleep quality (endogenous GH is primarily released during deep sleep)
- Anti-aging: improved skin elasticity, collagen, hair quality
- Increased bone density with long-term use

HOW IT WORKS

GH acts on GH receptors in the liver to stimulate IGF-1 production, which drives anabolic effects in muscle and bone. Directly, GH activates lipolysis via hormone-sensitive lipase in adipocytes and counteracts insulin's actions in fat tissue. GH also upregulates protein synthesis independently of IGF-1 via JAK2-STAT5 signaling in muscle.

DOSING

Anti-aging / TRT	Body recomp	Performance	Advanced split
1–2 IU/day	2–4 IU/day	4–6 IU/day	2–3x/day

PROTOCOLS

- **Anti-Aging / Wellness:** 1–2 IU SC daily before bed. Monitor IGF-1 every 3 months.

- **Body Recomp:** 2–3 IU SC daily (morning fasted or split AM/PM).
- **Performance:** 4–6 IU/day split into 2 injections (pre-training + before bed).
- **Cycle length:** Minimum 3 months for meaningful body composition changes; 6+ months optimal.

SIDE EFFECTS

- Water retention and edema (dose dependent — reduces over weeks)
- Carpal tunnel syndrome (tingling/numbness in hands — reduce dose if severe)
- Joint pain (arthralgias) — common early, usually transient
- Insulin resistance at higher doses — monitor fasting glucose

■ **Pharmaceutical GH is expensive (\$300–800+/month). Verify product via HPLC — counterfeit and underdosed GH is extremely common. Monitor IGF-1, fasting glucose, HbA1c every 3 months.**

Semax

Semax — ACTH(4-7)PGP Neuropeptide

Semax is a synthetic heptapeptide analogue of ACTH(4-7), developed by the Institute of Molecular Genetics (Russian Academy of Sciences) in the 1980s–90s. It is approved in Russia for stroke, TBI, cognitive decline, ADHD, and nervous system disorders. It dramatically increases BDNF (brain-derived neurotrophic factor) and NGF (nerve growth factor) — two of the brain's most important neuroplasticity molecules. Legal and widely used internationally as a research nootropic.

KEY BENEFITS

- Rapidly elevates BDNF and NGF — promotes neurogenesis and synaptic plasticity
- Enhances working memory, focus, and information processing speed
- Neuroprotective — shown effective in stroke, TBI, and ischemia recovery
- Reduces anxiety without sedation (anxiolytic without impairing cognition)
- Improves executive function, learning capacity, and attention
- Anti-fatigue — improves mental endurance and stamina under cognitive load
- Shown to improve ADHD symptoms without stimulant side effects

HOW IT WORKS

Semax is a melanocortin receptor (MCR) agonist that also directly stimulates BDNF and NGF gene expression in the brain. Its proline-glycine-proline (PGP) C-terminal extension makes it resistant to enzymatic degradation compared to native ACTH(4-7). It modulates the dopaminergic and serotonergic systems, and upregulates genes involved in synaptic transmission, neuroplasticity, and immune regulation in the CNS.

DOSING

SC injection	Starting dose	Frequency	Cycle pattern
200–600 mcg	100–200 mcg	1–2x daily	5–10 days on/off

PROTOCOLS

- **SC injection (standard):** 200–600 mcg SC, 1–2x daily. Inject abdomen or thigh.
- **Cycle protocol:** 5–10 days on, 5 days off to prevent receptor downregulation.
- **Cognitive enhancement:** 200 mcg SC 30 min before demanding work.
- **Recovery (TBI/stroke):** 600–900 mcg/day SC for 10–14 days under physician guidance.

SIDE EFFECTS

- Mild injection site irritation (transient)
- Mild headache at higher doses
- Transient fatigue after dose wears off (BDNF normalization)
- Rarely: mild mood elevation that can overshoot to slight agitation

■ *N-acetyl Semax (NA-Semax) and NA-Semax Amide are more potent variants — start at lower doses. Refrigeration required — peptide degrades rapidly at room temperature.*

Selank

Selank — Tuftsin Analogue Anxiolytic Peptide

Selank is a synthetic hexapeptide analogue of Tuftsin (a natural immunomodulatory tetrapeptide) developed by the Institute of Molecular Genetics, Russia. Approved in Russia for anxiety, depression, PTSD, and cognitive disorders. It produces anxiolytic effects comparable to benzodiazepines but without sedation, addiction potential, or cognitive impairment. Used clinically as a non-addictive anti-anxiety treatment.

KEY BENEFITS

- Potent anxiolytic — reduces anxiety equivalent to low-dose benzodiazepines
- Zero addiction potential — no GABA receptor downregulation
- No sedation — anxiety reduction without cognitive blunting
- Antidepressant effects — shown to reduce symptoms of generalized anxiety and depression
- Improves memory consolidation and learning speed
- Immunomodulatory — enhances immune function (Tuftsin mechanism)

- Reduces stress-induced cognitive dysfunction

HOW IT WORKS

Selank modulates GABA-A receptor sensitivity (without direct binding), reducing anxiety via the GABAergic system. It also increases BDNF, boosts serotonin uptake regulation, and modulates enkephalins (endorphin-like peptides). Unlike benzodiazepines, it does not bind GABA-A receptors directly — explaining the lack of sedation, tolerance, and dependence. It also upregulates IL-6 and immune regulatory genes.

DOSING

SC injection	Starting dose	Frequency	Typical cycle
250–500 mcg	100–200 mcg	1–3x daily	2–4 weeks

PROTOCOLS

- **Anxiety / Acute stress:** 250 mcg SC, 2x daily (AM + PM). Inject abdomen or thigh.
- **Daily nootropic:** 250–500 mcg SC morning.
- **Semax + Selank stack:** Semax (morning, cognitive drive) + Selank (evening, wind-down).
- **Cycle:** 14–21 days on, 7 days off — prevents adaptation.

SIDE EFFECTS

- Mild injection site irritation (transient)
- Very mild sedation possible at high doses (rare)
- No addiction, withdrawal, or physical dependence reported
- Mild headache (rare)

■ *N-acetyl Selank (NA-Selank) is more potent — reduce dose by 30–50%. Keep reconstituted solution refrigerated — degrades at room temperature within days.*

Oxytocin

Oxytocin — Neuropeptide Hormone

Oxytocin is a 9-amino-acid neuropeptide produced in the hypothalamus and released by the posterior pituitary. Known as the 'bonding hormone' or 'love hormone,' it is released during physical touch, orgasm, childbirth, and breastfeeding. In performance and wellness contexts, exogenous oxytocin is used to enhance sexual experience, improve emotional connection, reduce social anxiety, accelerate orgasm, and support post-workout recovery via anti-inflammatory pathways.

KEY BENEFITS

- Intensifies orgasm — facilitates and amplifies climax in men and women
- Increases emotional bonding and intimacy between partners
- Reduces social anxiety — improves confidence in social situations
- Enhances sexual arousal and physical sensitivity
- Anti-inflammatory — reduces cortisol and systemic inflammation post-exercise
- Accelerates wound healing (promotes myofibroblast activity)
- Improves trust, prosocial behavior, and relationship satisfaction

HOW IT WORKS

Oxytocin binds oxytocin receptors (OTR) throughout the brain and body. In the limbic system, it activates reward circuits (dopamine release) and reduces amygdala reactivity to threatening stimuli — explaining its anxiolytic and bonding effects. In sexual function, it amplifies genital sensation and facilitates spinal ejaculatory/orgasm reflexes. Peripherally, it reduces inflammation via NF-κB pathway inhibition.

DOSING

SC injection	Standard dose	Timing	Frequency
10–40 IU	20 IU	20–30 min prior	As needed

PROTOCOLS

- **Sexual enhancement:** 20–40 IU SC injection 20–30 min before intimacy.
- **Social anxiety:** 20 IU SC 30–45 min before social event.
- **Post-workout recovery:** 20 IU SC post-training for anti-inflammatory effect.
- **Orgasm enhancement:** 20 IU for women 30 min pre-activity; pair with PT-141 for enhanced desire + intensity.

SIDE EFFECTS

- Mild injection site irritation (transient)
- Mild nausea at higher doses (>40 IU)
- Possible oxytocin 'crash' — emotional low after peak (dose dependent)
- In social contexts: can amplify negative emotions as well as positive (caution in contentious situations)

■ **Contraindicated during pregnancy — causes uterine contractions. Effects are bidirectional on emotion — may amplify negative social experiences as well as positive ones.**

SLU-PP-332

SLU-PP-332 — ERRα/γ Agonist (Exercise Mimetic)

SLU-PP-332 is a synthetic agonist of estrogen-related receptors alpha and gamma (ERRα and ERRγ) — nuclear receptors that regulate the expression of genes controlling mitochondrial biogenesis, oxidative metabolism, and energy expenditure. Published in the Journal of Medicinal Chemistry (2023), SLU-PP-332 increased endurance in mice by 70% and promoted significant fat loss independent of diet or exercise. Among the most compelling emerging exercise mimetics in preclinical research.

KEY BENEFITS

- 70% increase in treadmill endurance capacity in preclinical studies (without training)
- Significant fat loss — promotes oxidative fat metabolism in skeletal muscle
- Mitochondrial biogenesis — increases mitochondrial density in muscle tissue
- Shifts muscle fiber type toward slow-twitch / oxidative (endurance-type)
- Reduces cardiac hypertrophy and improves heart failure markers in animal models
- Exercise mimetic — produces aerobic training adaptations without exercise
- Potential application in obesity, type 2 diabetes, heart failure, sarcopenia

HOW IT WORKS

SLU-PP-332 activates ERRα and ERRγ — orphan nuclear receptors that normally respond to energy stress signals (like those generated by exercise). Activation of these receptors triggers expression of PGC-1α target genes — the master regulator of mitochondrial biogenesis. This results in increased mitochondrial number, enhanced fatty acid oxidation, muscle fiber type shifting, and improved cardiac function, all mimicking the cellular adaptations of sustained aerobic training.

DOSING

No human trials yet	Preclinical dose	Extrapolated human	Route (unclear)
N/A (human)	Mouse: 30 mg/kg	~20–50 mg?	Daily SC/oral

PROTOCOLS

- **IMPORTANT:** No established human protocol — all current use is self-experimentation.
- **Extrapolated from mice:** ~20–50 mg daily oral or SC (highly uncertain).
- **Stack context:** Often grouped with MOTS-c, BAM-15, 5-Amino-1MQ for metabolic/exercise mimetic protocols.
- **Monitoring:** Track cardiovascular markers, liver enzymes, and body composition changes.

SIDE EFFECTS

- Unknown in humans — no human clinical trial data available
- Preclinical: no significant toxicity observed at study doses
- Theoretical: ERR activation in cancer cells could promote proliferation (monitor)
- Unknown cardiovascular, renal, and hepatic effects in humans long-term

■ **Extreme caution — zero human clinical trial data. Among the most experimental compounds on this list. Use only with thorough understanding of the risk profile.**

BAM-15

BAM-15 — Mitochondrial Protonophore Uncoupler

BAM-15 is a next-generation mitochondrial protonophore — a mitochondrial uncoupling agent that 'short-circuits' the electron transport chain, causing energy (calories) to be released as heat instead of being stored as ATP. Unlike DNP (2,4-dinitrophenol), BAM-15 acts only on the inner mitochondrial membrane and does not dissipate the plasma membrane electrochemical gradient — making it substantially safer in preclinical models. Published in Nature Communications (2020) — significant fat loss without diet change in mice.

KEY BENEFITS

- Significant fat loss by converting stored energy to heat (thermogenesis)
- Preserves lean muscle mass during fat loss — unlike most caloric-restriction approaches
- Improves insulin sensitivity — reduces insulin resistance in obese mouse models
- Reduces liver fat (hepatic steatosis) and systemic inflammation
- No suppression of appetite — mechanism is purely thermogenic
- Improved mitochondrial efficiency markers
- Synergistic with exercise — enhances the uncoupling that occurs naturally during training

HOW IT WORKS

BAM-15 is a proton carrier — it shuttles protons (H+) across the inner mitochondrial membrane down the electrochemical gradient, bypassing ATP synthase. This 'uncouples' oxidative phosphorylation from ATP production, so the energy from fuel oxidation is released as heat. The cell compensates by burning more fuel — creating a caloric deficit at rest. BAM-15 specificity for the mitochondrial membrane prevents the dangerous plasma membrane depolarization that makes DNP lethal at higher doses.

DOSING

No human trials	Preclinical dose	Likely route	Status
Unknown (human)	Mouse: 50 mg/kg	Oral	Research only

PROTOCOLS

- **CRITICAL:** No human dose established — BAM-15 is pure preclinical research compound.
- **Animal studies:** 50 mg/kg/day oral in mice — human equivalent roughly 4 mg/kg (HED).
- **Current status:** Not available from mainstream peptide vendors; strictly research compound.
- **If experimenting:** Cardiovascular monitoring (body temp, heart rate, blood pressure) is essential.

SIDE EFFECTS

- Unknown in humans — no human safety data exists
- Preclinical: no significant body temperature elevation at study doses (unlike DNP)
- Theoretical: hyperthermia, tachycardia, sweating at high doses (uncoupler class effect)
- Theoretical: mitochondrial dysfunction if mechanism is overwhelmed

■ ■ **EXTREME CAUTION** — *Research compound with zero human safety data. DNP, a related uncoupler, has caused multiple deaths. Do not extrapolate DNP experience to BAM-15. For research monitoring only.*

Cagrilintide

Cagrilintide (CAG) — Long-Acting Amylin Analogue

Cagrilintide is a long-acting synthetic analogue of amylin — a pancreatic hormone co-secreted with insulin that signals satiety and slows gastric emptying. Developed by Novo Nordisk, it has a ~7-day half-life (once-weekly dosing) and is being studied in Phase 3 trials as CagriSema (cagrilintide 2.4 mg + semaglutide 2.4 mg). REDEFINE 1 trial data showed ~25% body weight reduction — potentially surpassing every approved weight loss drug on the market. Acts via a completely different receptor than GLP-1 agonists.

KEY BENEFITS

- ~25% body weight reduction in REDEFINE 1 trial (CagriSema combination at 68 weeks)
- Amylin receptor activation — entirely different pathway from GLP-1 agonists
- Powerful satiety signaling — reduces meal size and caloric intake
- Slows gastric emptying independently of GLP-1 mechanism
- Synergistic with semaglutide/tirzepatide — additive weight loss when combined
- Reduces postprandial glucose excursions
- Improves metabolic markers: lipids, blood pressure, HbA1c

HOW IT WORKS

Cagrilintide is an amylin analogue — it binds amylin receptors (which are calcitonin receptor/RAMP complexes) in the area postrema and NTS (nucleus tractus solitarius) in the brainstem, as well as in the hypothalamus. This produces satiety signals independent of GLP-1R activation. The combination with semaglutide provides dual central + peripheral satiety and gastric slowing — explaining the additive weight loss seen in CagriSema trials that exceeds either drug alone.

DOSING

Starting dose	Target dose	Frequency	Ramp-up
0.16 mg/wk	2.4 mg/wk	Once weekly	16-week titration

PROTOCOLS

- **CagriSema titration:** 0.16 mg x4 wks → 0.35 mg x4 → 0.7 mg x4 → 1.4 mg x4 → 2.4 mg.
- **Monotherapy:** Same titration schedule — studied as standalone in Phase 2.
- **Combination:** Pair with semaglutide 2.4 mg/week for maximum fat loss (CagriSema protocol).
- **Protein/exercise:** 1g/lb protein + resistance training mandatory to preserve lean mass.

SIDE EFFECTS

- Nausea (common, GI class effect — diminishes with titration)
- Injection site reactions (mild erythema, nodules at injection site more common than with GLP-1 alone)
- Constipation, diarrhea
- Hypoglycemia (in combination with insulin or sulfonylureas)

■ **Not yet FDA approved as of 2025 — NDA submission expected 2025–2026. Injection site nodules are more common than with semaglutide alone — rotate sites consistently.**

MEDICAL DISCLAIMER: This guide is for educational purposes only. None of this constitutes medical advice. Peptides and research compounds described may not be FDA approved for the uses described. Always consult a qualified physician before using any compound. Verify supplier quality via third-party HPLC testing. Do not use any compound without understanding its full risk profile. forgedbyfreedom.com